# Sahaj Bhadja

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## Summary

Product-focused Software Engineer with 2+ years building **full-stack AI applications** using **TypeScript, React, Python, LLM APIs**, owning customer-facing workflows from product requirements and interface design through testing and production.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent product usage.
- Designed multi-tenant storage with **row-level security** for 500+ users, cutting dashboard latency from 1.2s to 350ms while shipping customer-facing workflows through CI/CD at 99%+ reliability across production releases.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Built **React and Flask product workflows** for uploads, grading history, and feedback, serving 150+ instructors and 10K+ daily requests while incorporating user requirements into reliable customer-facing features.
- Created a **unified AI model layer** across OpenAI, Anthropic, and Qwen with vision fallbacks, timeout handling, usage tracking, and automated tests, reducing customer feedback turnaround 60% for production reliability.

**USC AD Labs** **January 2025 – May 2025**
*ML Research Assistant* *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** with data pipelines over 100K+ trajectory logs, improving long-horizon planning 18% while translating research objectives into reproducible model experiments with measurable outcomes.
- Designed a multi-head **attention** encoder fusing map geometry with agent trajectories, comparing three baselines through documented evaluations and improving closed-loop stability 15% across iterative research cycles with consistent results.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern – VibrantChem* *India*

- Built **React and Node.js product workflows** for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records across locations through REST APIs and real-time WebSocket updates reliably at scale.
- Owned eight services from implementation through deployment, adding **automated testing, monitoring, Docker, and Jenkins CI/CD** that cut release time 50% while sustaining 95%+ reliability for customer workflows.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Built **React and TypeScript customer workflows** for booking, pricing, coupons, and payment-status updates, serving 50K+ records through Flask APIs with sub-200-millisecond responses and real-time synchronization for customers daily.
- Mapped 12+ integrations and presented **end-to-end product designs** to clients before implementation, incorporating stakeholder feedback to reduce design errors 25% while delivering payment and notification workflows across delivery cycles.

## Skills

**Languages:** TypeScript, JavaScript, Python, Java, SQL, HTML5, CSS3, C++
**Frontend & Product:** React.js, Next.js, TailwindCSS, Responsive Design, Product Design, User Workflows, Figma, Accessibility
**AI & ML:** LLM APIs, GPT-4o, Claude, AI Agents, Prompt Engineering, RAG, Embeddings, FAISS, PyTorch, Transformers, NLP
**Backend & Data:** FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, MySQL, MongoDB, Redis, Authentication, Authorization
**Cloud & Engineering:** AWS, Azure, GCP, Docker, Kubernetes, Git, CI/CD, Jenkins, PyTest, Jest, Monitoring, Agile, Customer Collaboration

## Projects

**ResearchStudio** | *Next.js, TypeScript, FastAPI, PostgreSQL, WebSockets, LLM APIs*

- Designed a **full-stack customer-research workspace** for creating studies, managing participant sessions, streaming interview progress, and reviewing structured findings through responsive Next.js interfaces, FastAPI services, and PostgreSQL workflows.

**InsightSynthesis Engine** | *Python, FastAPI, LLM APIs, Embeddings, Redis, PostgreSQL*

- Designed an **AI insight-synthesis pipeline** that processes interview transcripts, extracts themes and supporting evidence, groups semantically related responses, and exposes traceable summaries through asynchronous FastAPI jobs and reviewable outputs.

**Big-Brain** Live | *LLaMA-2, FAISS, Pinecone, Clerk, PostgreSQL, GCP*

- Built an **AI application** with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents with Clerk authentication, cutting retrieval latency 40% at 95% query accuracy in production.